

Title: Explainable Machine Learning Framework for Early Heart Disease Prediction

Abstract

Heart disease is one of the leading causes of mortality worldwide, making early prediction and diagnosis essential for improving patient outcomes and reducing healthcare costs. Traditional machine learning models can accurately predict the presence of heart disease but often function as black-box systems, making it difficult for healthcare professionals to understand the reasoning behind their predictions. This project proposes an **Explainable Machine Learning Framework for Early Heart Disease Prediction** that combines powerful machine learning algorithms with Explainable Artificial Intelligence (XAI) techniques to provide accurate predictions along with transparent and interpretable explanations.

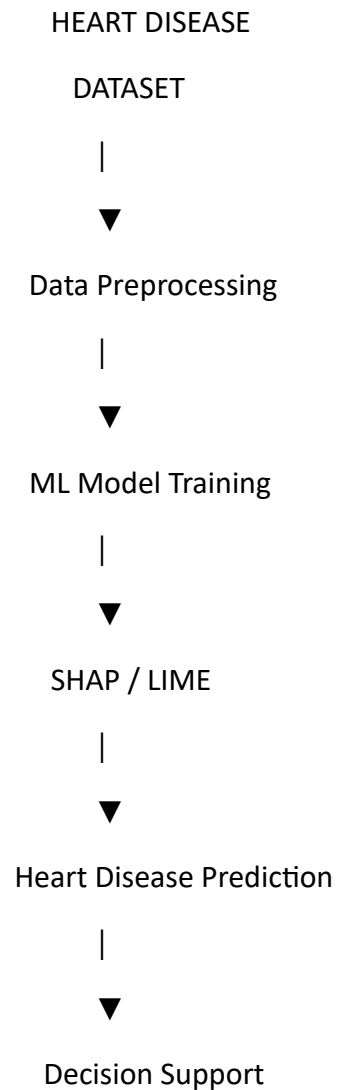
The proposed framework utilizes publicly available healthcare datasets such as the UCI Heart Disease Dataset, Cleveland Heart Disease Dataset, and Kaggle Heart Disease Dataset. The system performs data preprocessing, including missing value handling, feature selection, normalization, and class balancing before training machine learning models such as **Random Forest, XGBoost, Support Vector Machine (SVM), Logistic Regression, and Decision Tree**. The best-performing model is selected based on evaluation metrics. To improve transparency and trust, explainability techniques such as **SHAP (SHapley Additive Explanations)**, **LIME (Local Interpretable Model-Agnostic Explanations)**, and **feature importance analysis** are incorporated to identify the medical attributes that contribute most significantly to the prediction, such as age, cholesterol level, blood pressure, chest pain type, maximum heart rate, and fasting blood sugar.

The proposed framework further assists healthcare professionals by providing interpretable visual explanations that support clinical decision-making and enable better understanding of patient risk factors. The performance of the prediction model is evaluated using **accuracy, precision, recall, F1-score, ROC-AUC score, and confusion matrix**, while the explainability component is assessed based on the consistency and reliability of generated explanations. The resulting system aims to deliver not only highly accurate early heart disease prediction but also trustworthy and transparent insights that can support preventive healthcare, timely diagnosis, and improved patient care.

Datasets

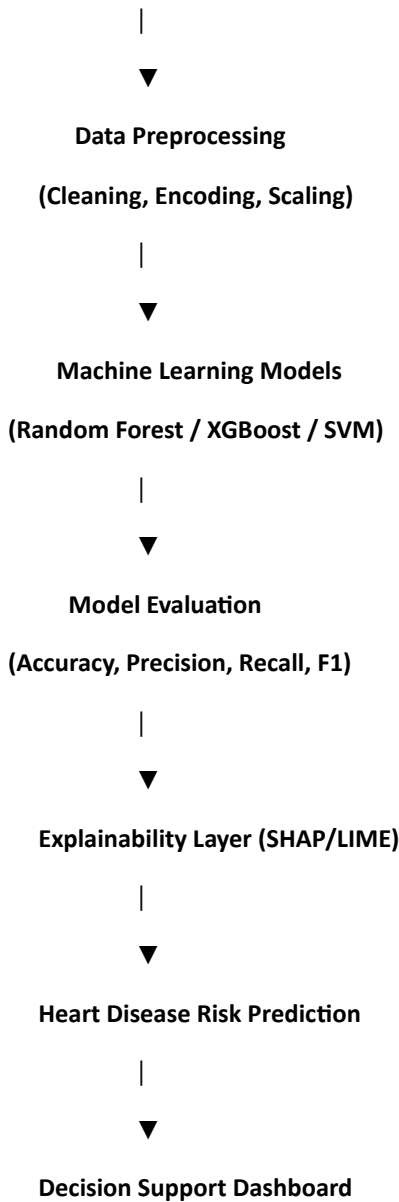
Dataset	URL Link
UCI Heart Disease Dataset	https://archive.ics.uci.edu/ml/datasets/heart+disease
Cleveland Heart Disease Dataset	https://www.kaggle.com/datasets/cherngs/heart-disease-cleveland-uci
Heart Disease Dataset	https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

Flowchart



Approach

HEART DISEASE DATASETS



Project Group Members

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